Volcano for the Agentic AI Era

Unified Scheduling Across Training, Inference, and Agents

KubeCon + CloudNativeCon North America 2026

SPEAKERS / AFFILIATIONS — TO CONFIRM

待确认：讲者姓名、署名顺序与单位；当前为工作版式，未套用官方模板。

Different Workloads, Shared Infrastructure

Different arrival patterns

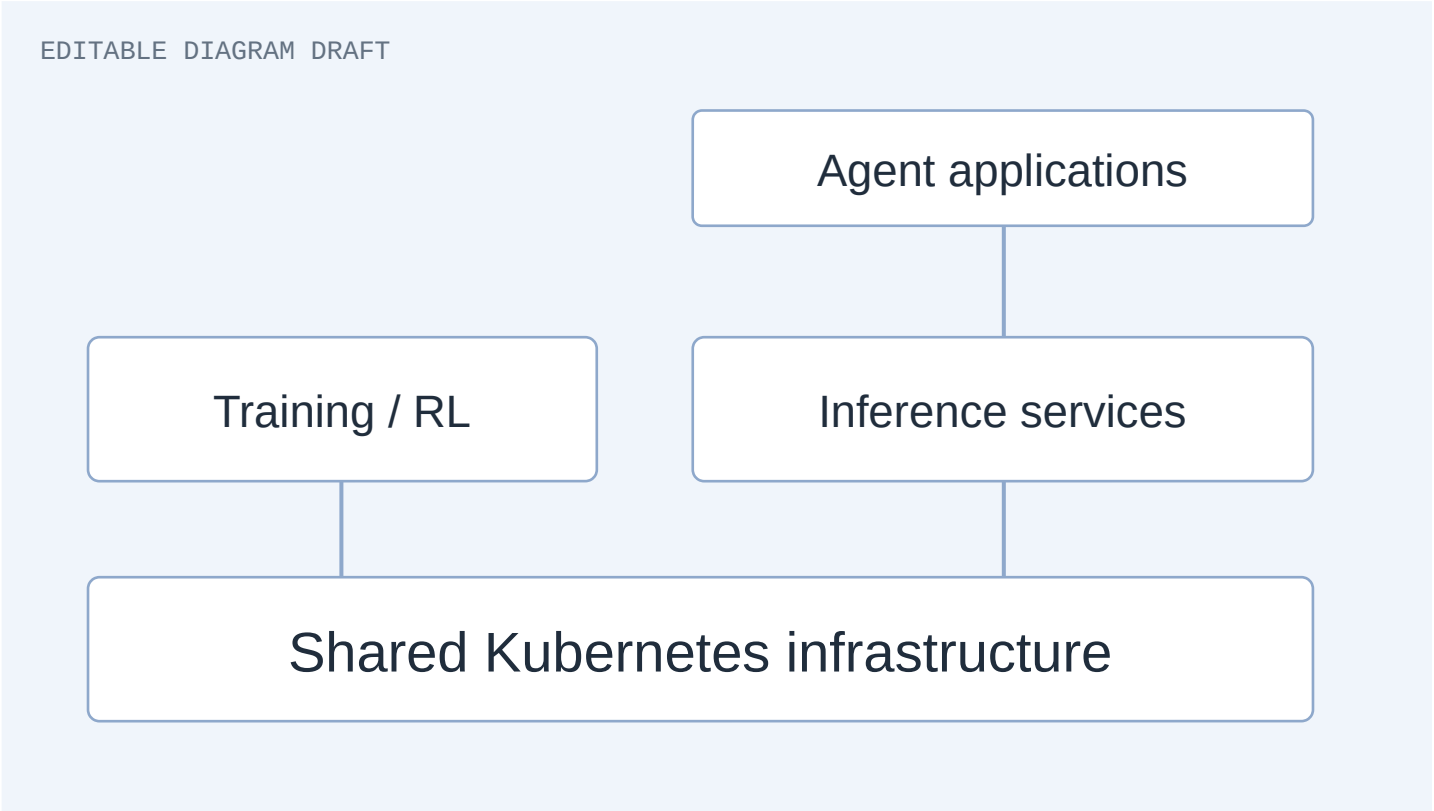
Long-running services, grouped compute, and bursts.

Different scheduling needs

Responsiveness, complete groups, and communication locality.

A shared resource pool

Coordinate infrastructure without forcing one execution model.



复用：生命周期图精简。 Agent 应用依赖推理服务，不把每次调用画成一个新 Pod 。

A Coordinated Scheduling Framework

Keep different scheduling paths

Support distinct workload requirements.

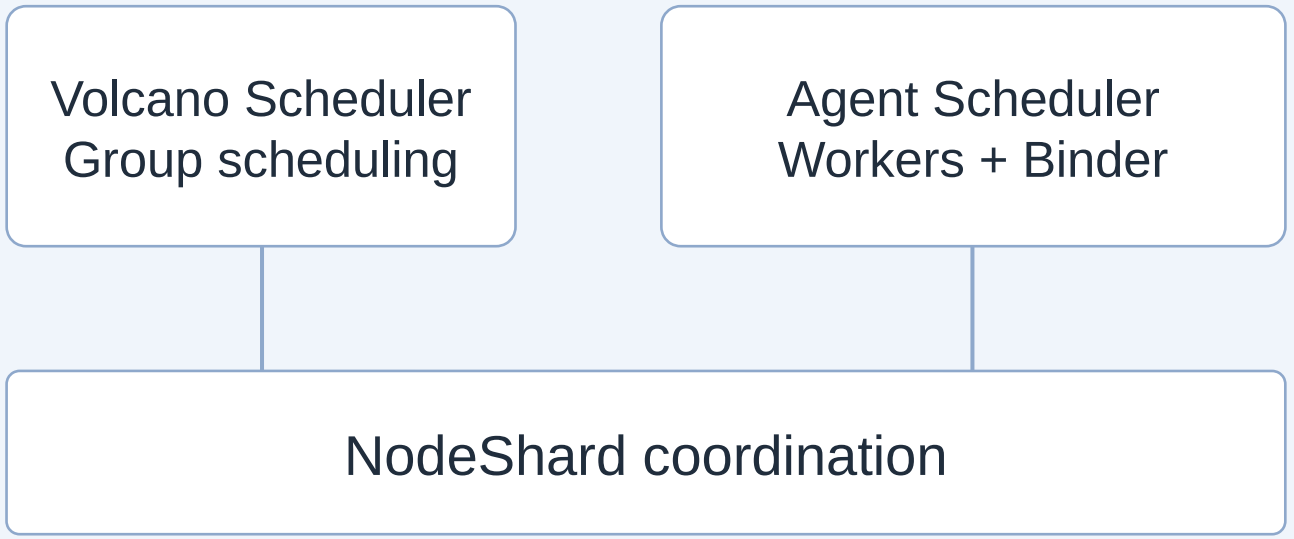
Coordinate access to nodes

NodeShard connects the paths at the infrastructure layer.

Handle local conflicts

Workers and Binder coordinate within the Agent Scheduler.

SCHEMATIC — REUSE / SIMPLIFY EXISTING ARCHITECTURE



Scheduler-internal conflicts ≠ cross-scheduler coordination

复用： China 7–8 / EU 9 。 worker / Binder 讲 30–45 秒；本页不放性能提升数字。

NodeShard: Coordinating Node Access

Controller sets desired membership

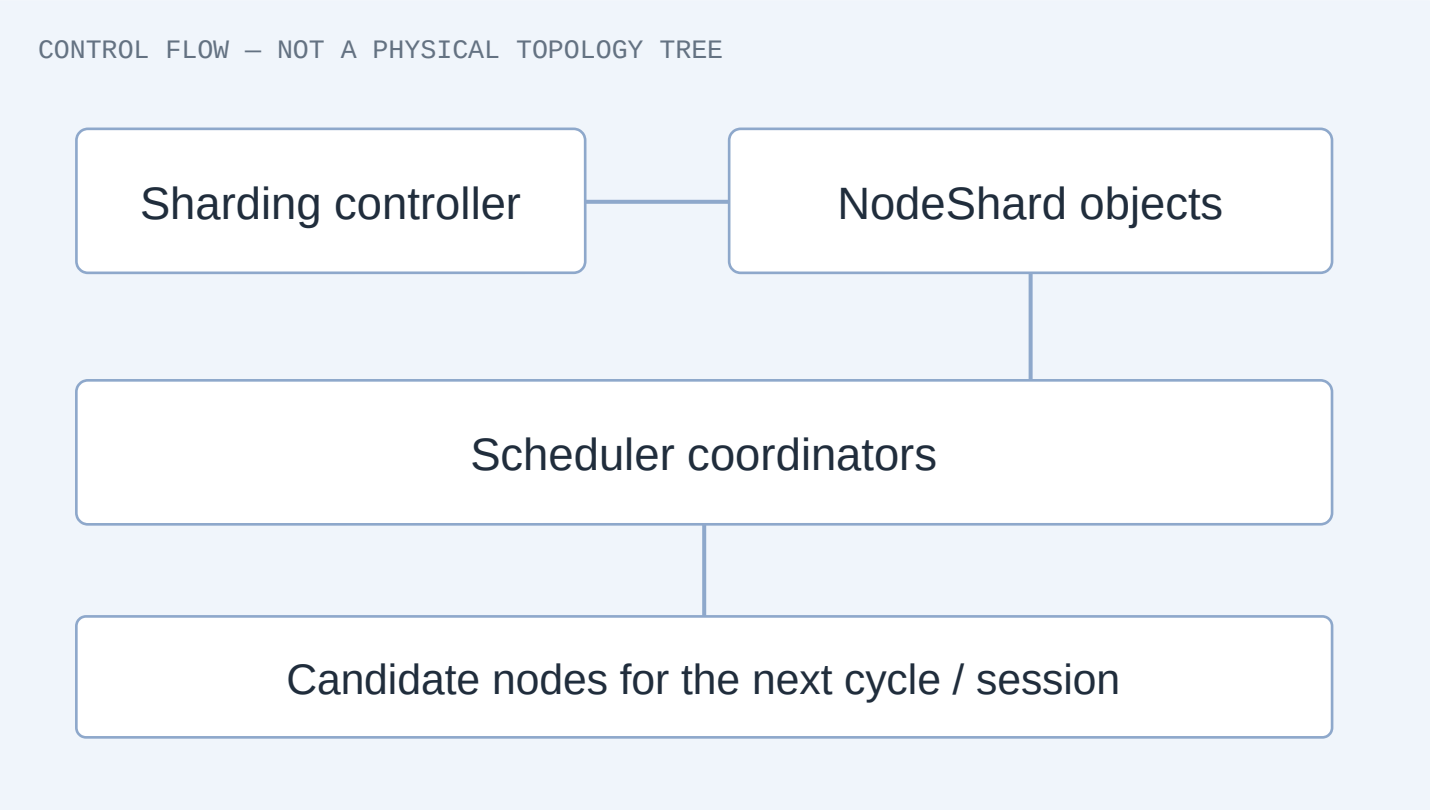
Express which nodes belong in each desired shard.

Coordinators synchronize usage

Reconcile desired and in-use nodes around scheduling cycles.

Mode defines the boundary

none: all nodes · soft: preference · hard: restricted scope



复用： China 9 / EU 10–11 。补一个节点范围交接例子；不是迁移运行中的 Pods 。

Available Nodes Are Only the Beginning

Enough free resources can still leave a workload unable to run.

01

Complete replicas

Do all required members fit?

02

Communication locality

Are the nodes in the right domain?

03

Group relationships

Should instances share a domain?

新增转场： NodeShard 是候选范围， HyperNode 是物理拓扑；不要画成父子层级。

PodGroup and SubGroup: Complete Replicas

PodGroup: overall gang requirements

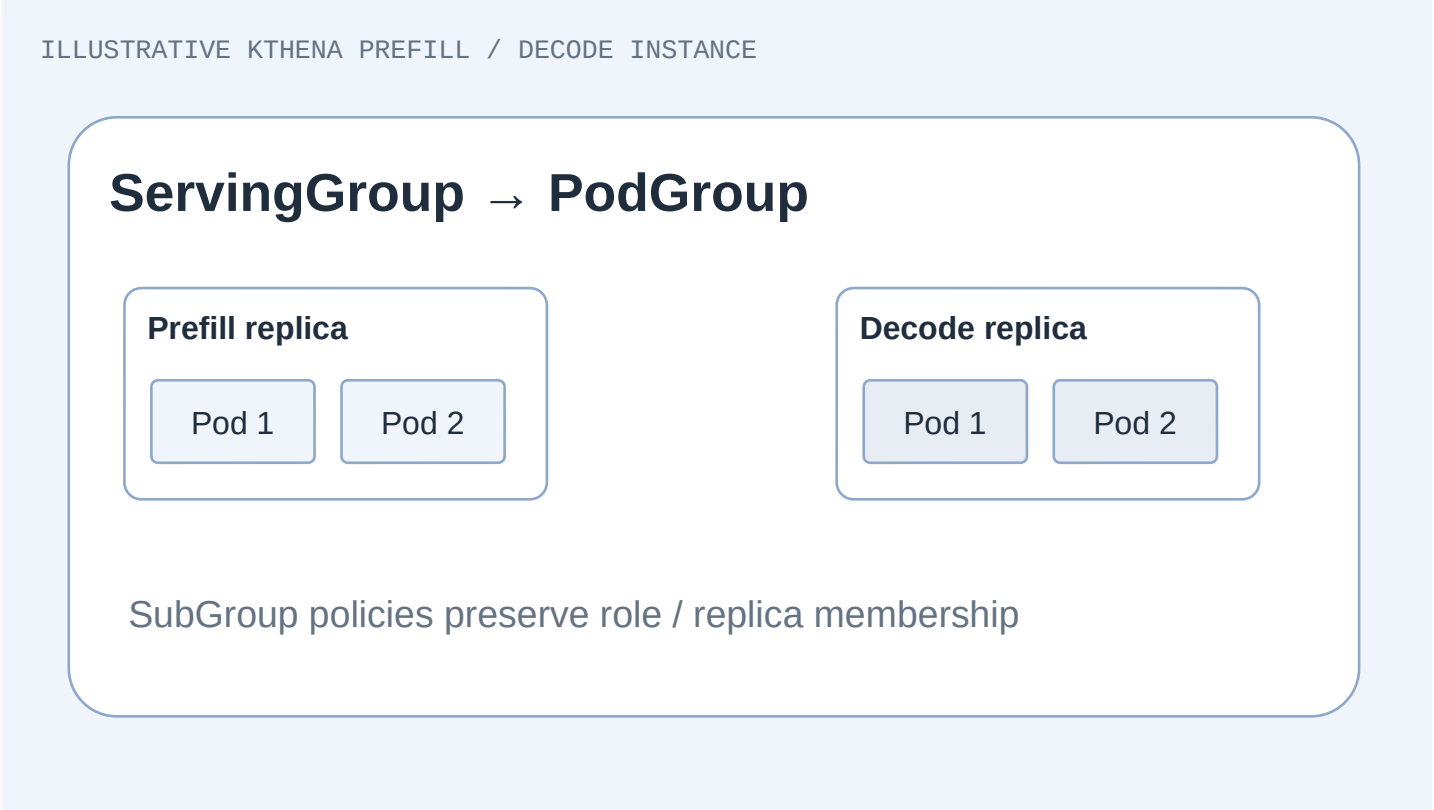
Describe what the workload needs as a whole.

SubGroup: complete member sets

Preserve a replica that spans multiple Pods.

Use role and replica identity

Extra Pods in one role cannot fill a missing replica in another.



复用： China 14 、 22 。 policy 可以产生多个实际 SubGroups ； 以下为 P/D 示意，不是完整 API 。

HyperNode: Expressing Topology Domains

Describe nested infrastructure domains

Represent nodes and their topology hierarchy.

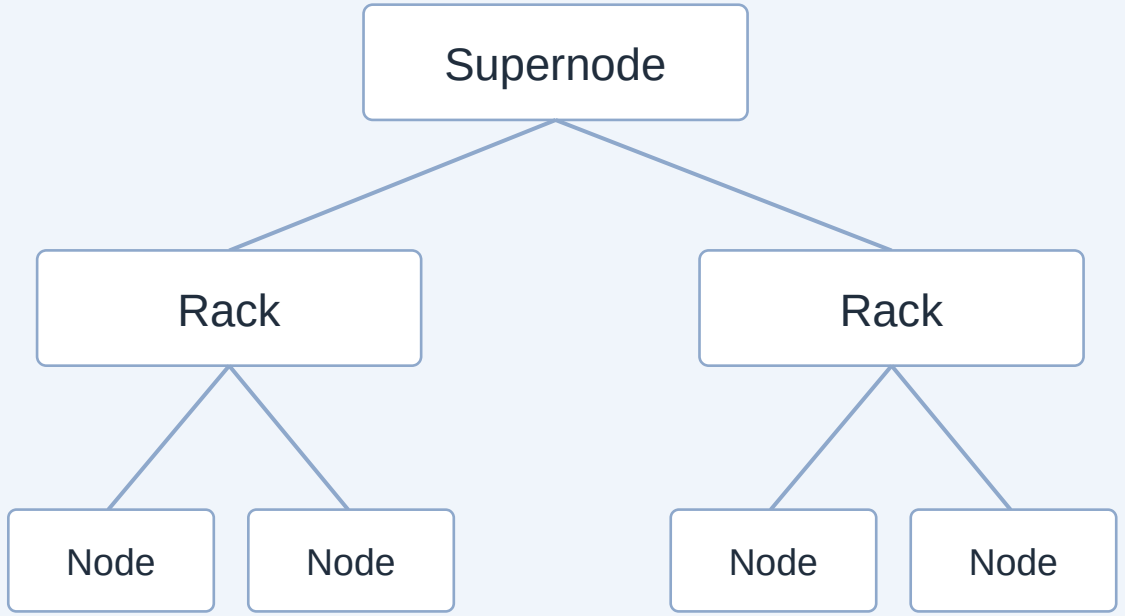
Choose a placement level

A group can target an appropriate domain size.

Express hard or soft constraints

State whether locality is required or preferred.

EDITABLE DIAGRAM DRAFT



复用： China 16 。仅保留一棵树；层级名称按现场集群映射，不写固定带宽收益。

Place Groups at Different Topology Levels

One boundary for the instance

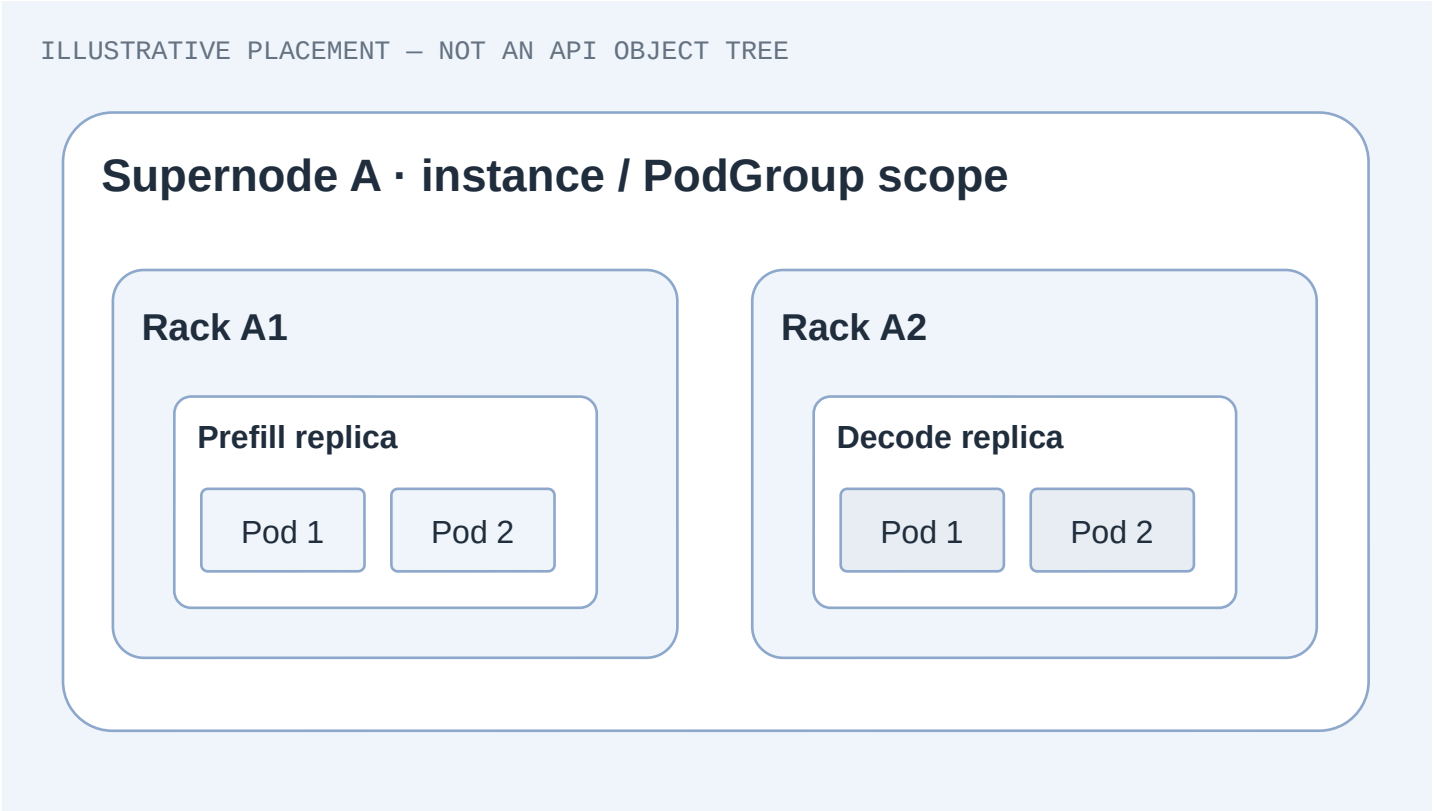
Keep the PodGroup inside a larger topology domain.

Tighter boundaries for replicas

Place each multi-Pod replica in a smaller domain.

Check capacity within the domain

Cluster-wide capacity alone is insufficient.



复用： China 17 。延续 P/D 案例；用两层边框区分实例范围和完整副本范围。

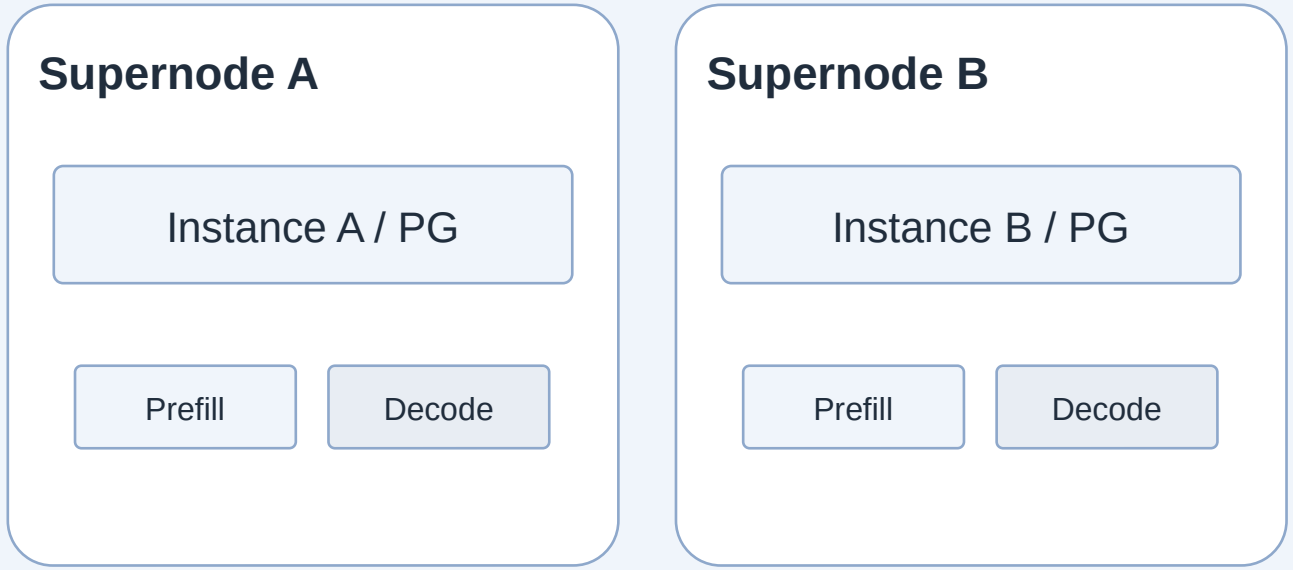
Keep Instances Together, Place Them Apart

AVAILABLE IN v1.16

Locality within.
Separation between.

Combine PodGroup network topology with anti-affinity across selected PodGroups.

EDITABLE DIAGRAM DRAFT



Within-group topology + between-group anti-affinity

SOURCE Volcano v1.16 · PodGroup anti-affinity #5349

新增重点： v1.16 PodGroup anti-affinity 。复用 China 15 的构图，更新功能状态。

Controllers Bring Workload Structure to Scheduling

PROJECT	EXISTING / MERGED	NEXT / IN PROGRESS
Kthena	Role / replica gang + topology	Reference use cases for group relationships
KubeRay	SubGroup support · #5249 merged	Replica topology · #5369 open
Grove	Gang / SubGroup backend merged	Topology mapping not yet supported
Kubeflow	Existing Volcano integration	Finer TrainJob groups + topology
LWS	Existing gang / Volcano integration	Explore roles / slices + group relationships

Controllers supply identity; the scheduler applies group and placement constraints.

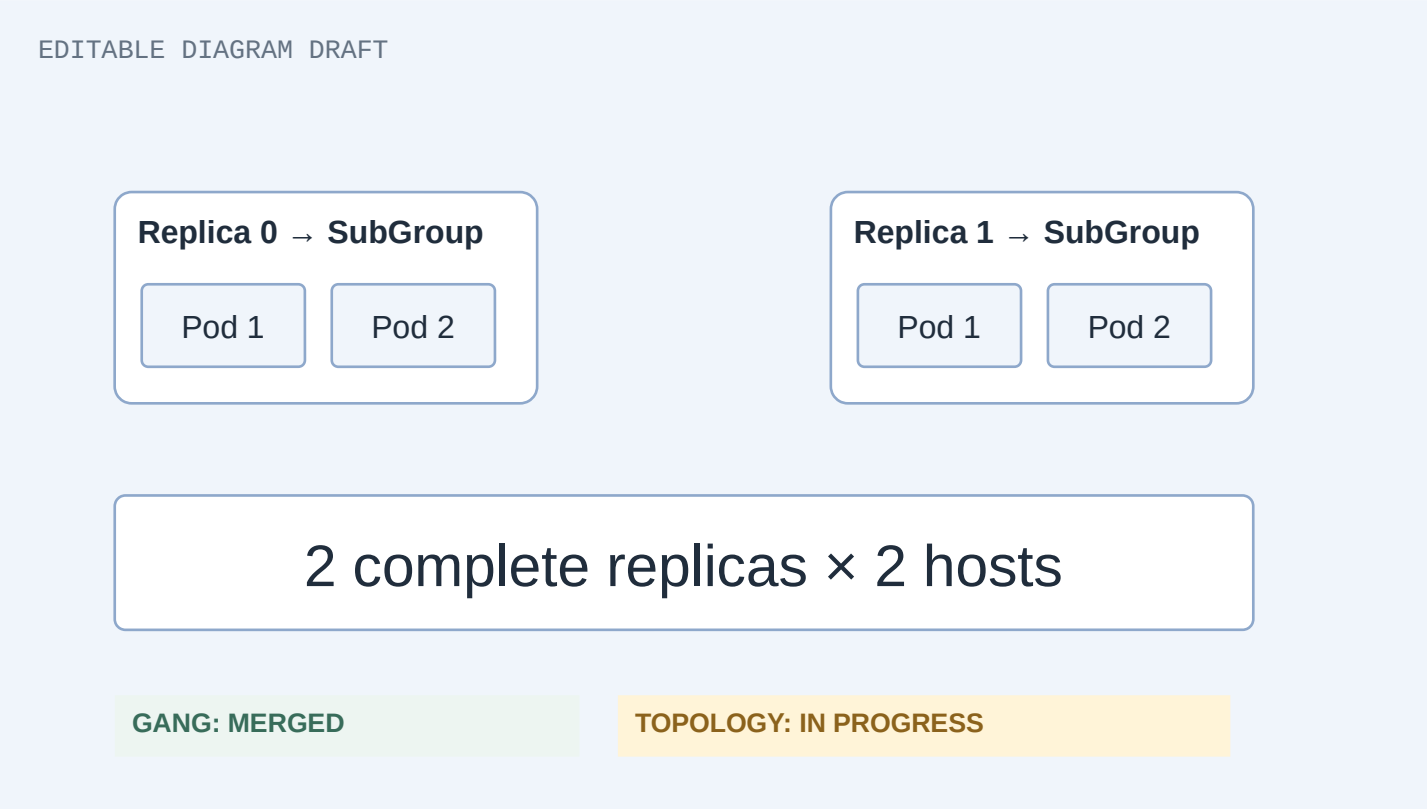
状态截至 2026-10-28；已合并不等于下游稳定版已发布。演讲前核对 PR 和版本。

KubeRay: A Replica Can Span Multiple Pods

```
minReplicas = 2
numOfHosts = 2
```

The requirement is two complete replicas, each spanning two Pods.

Any four Pods are not enough.



新增案例： WorkerGroup / policy / 实际 replica 不混为一谈；省略 head ，仅聚焦一个 worker group 。

Extending the Ecosystem and Placement Model

Available foundation

Standalone HyperNode controller

A reusable topology API

Integration work

KubeRay / Grove topology

Deeper Kubeflow / LWS mappings

Proposed semantics

SubGroup affinity / anti-affinity

Within the same PodGroup

As integrations grow, we need repeatable ways to evaluate scheduling behavior.

Standalone HyperNode 已交付； SubGroup 关系与进一步的 controller / adapter 接入分别标状态。

A Repeatable Scheduling Benchmark Framework

Choose an environment

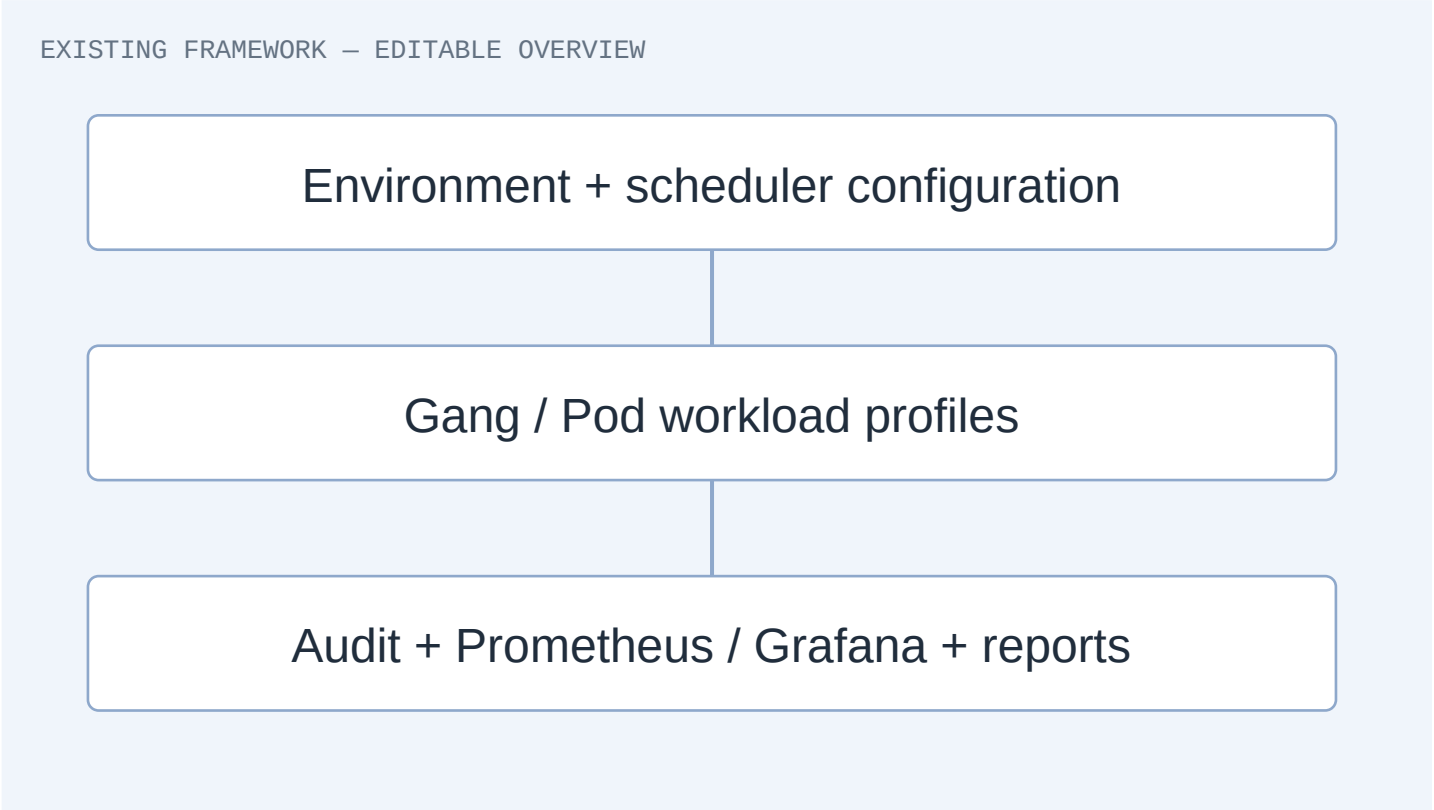
Kind + KWOK, or an existing Kubernetes cluster.

Choose a workload profile

Gang jobs, bare Pods, and topology configurations.

Collect observations

Audit events, scheduler metrics, dashboards, and reports.



基于仓库 benchmark/ 。框架能力与具体测试结果分开； KWOK 不模拟真实 GPU 计算和网络性能。

Progress So Far—and What We Are Still Testing

IN THE FRAMEWORK

Configurable test scenarios

Gang / Pod profiles and topology setup

Latency and throughput collection

Audit-based observations and scheduler metrics

Repeatable setup and reporting

Scripts, dashboards, and saved reports

VALIDATION IN PROGRESS

Representative workload shapes

Scale, job sizes, and topology constraints

Optimization comparisons

Matched configurations and repeated runs

Measurement boundaries

Separate creation, scheduling, and app readiness

进展页：左侧是已核查的框架能力；右侧是待讲者确认的测试重点，不写成已完成结果。

Benchmark Snapshot: Results Under Validation

WORK IN PROGRESS

One scenario. One question.

Show what changed, what was measured, and what remains under test.

Freeze the claim after validation.

RESULT PLACEHOLDER — NO MEASURED DATA YET

Insert one verified chart or a real dashboard capture

Build / cluster / nodes / workload: TBD

Metric / window / repetitions: TBD

Observation / limits / next test: TBD

待填：选择一项正在复测的结果，补环境、版本、负载与重复次数。当前没有放入测量数据。

Framework, Workloads, Ecosystem, Evidence

01

Coordinate scheduling paths

Dual paths and NodeShard coordination

02

Preserve workload structure

Multi-level gang and network topology

03

Extend ecosystem integrations

Existing adoption and deeper controller mappings

04

Measure and improve

Repeatable benchmarks and ongoing validation

Contribute workload profiles, reproducible observations, and integration feedback.

收尾：测试和优化仍在继续，邀请贡献真实 workload 与 benchmark profile；不把 WIP 写成完成。

Questions?

What should your workload keep together—or place apart?

Scheduling paths
NodeShard coordination

Workload groups
Gang + topology + relationships

github.com/volcano-sh/volcano

备份材料： NodeShard 模式、关键 API 、生态版本与 benchmark 方法；正式版移除此提示。